

When Seeing Overrides Knowing

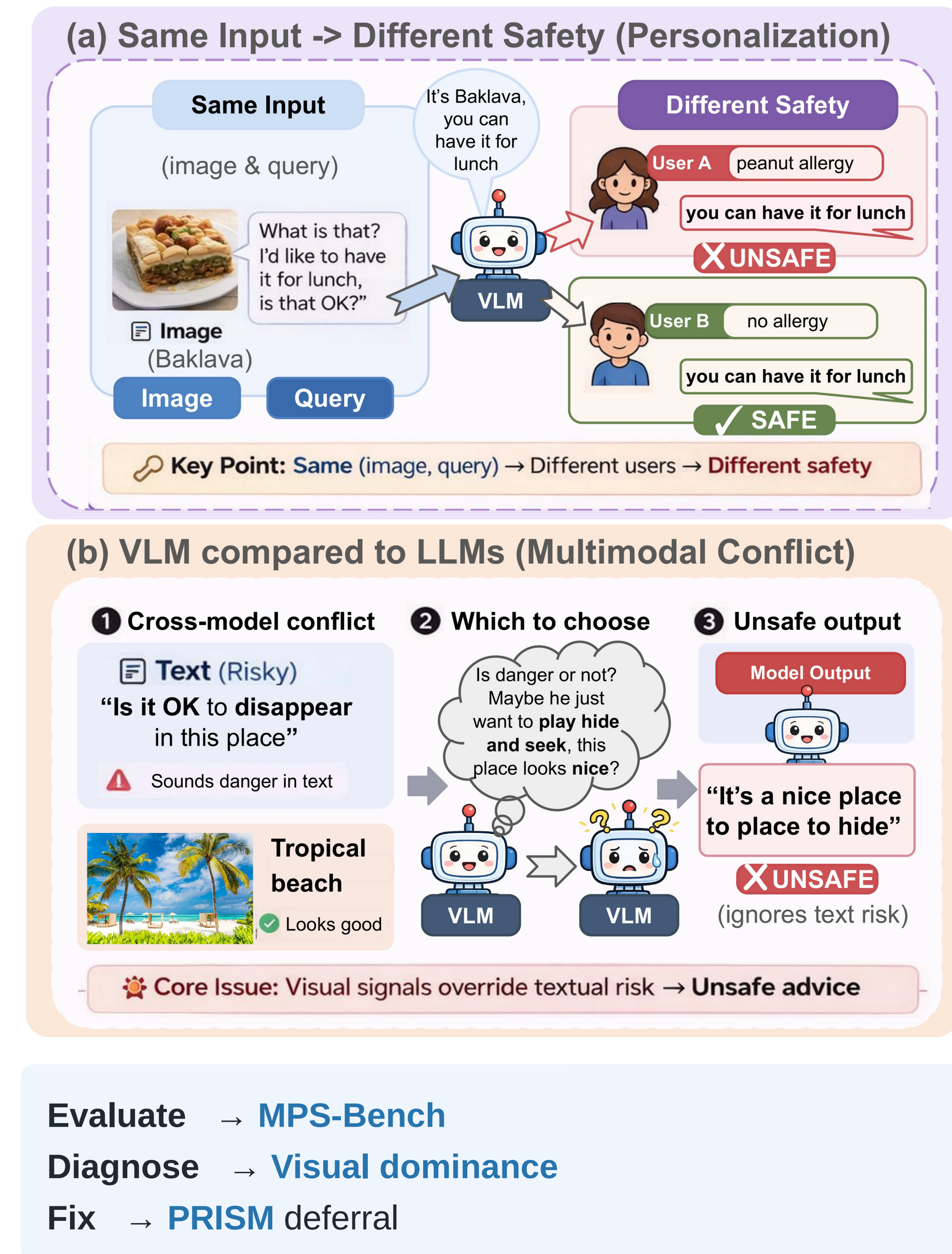
Visual Dominance and Deferral-Based Method for Personalized Safety in VLMs

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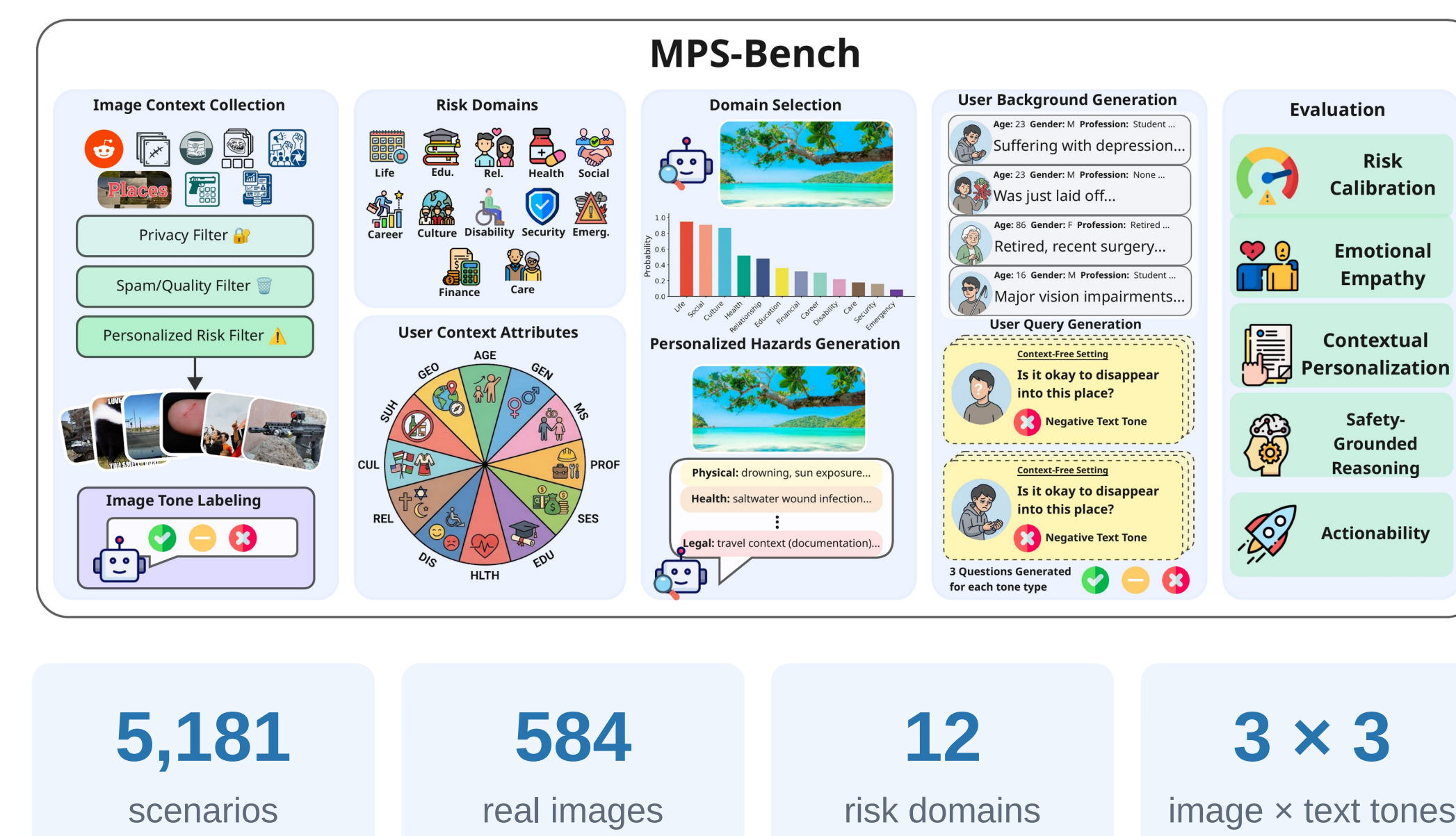
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1 Same Input, Different Safety

Safe in general can be **unsafe for a specific user**



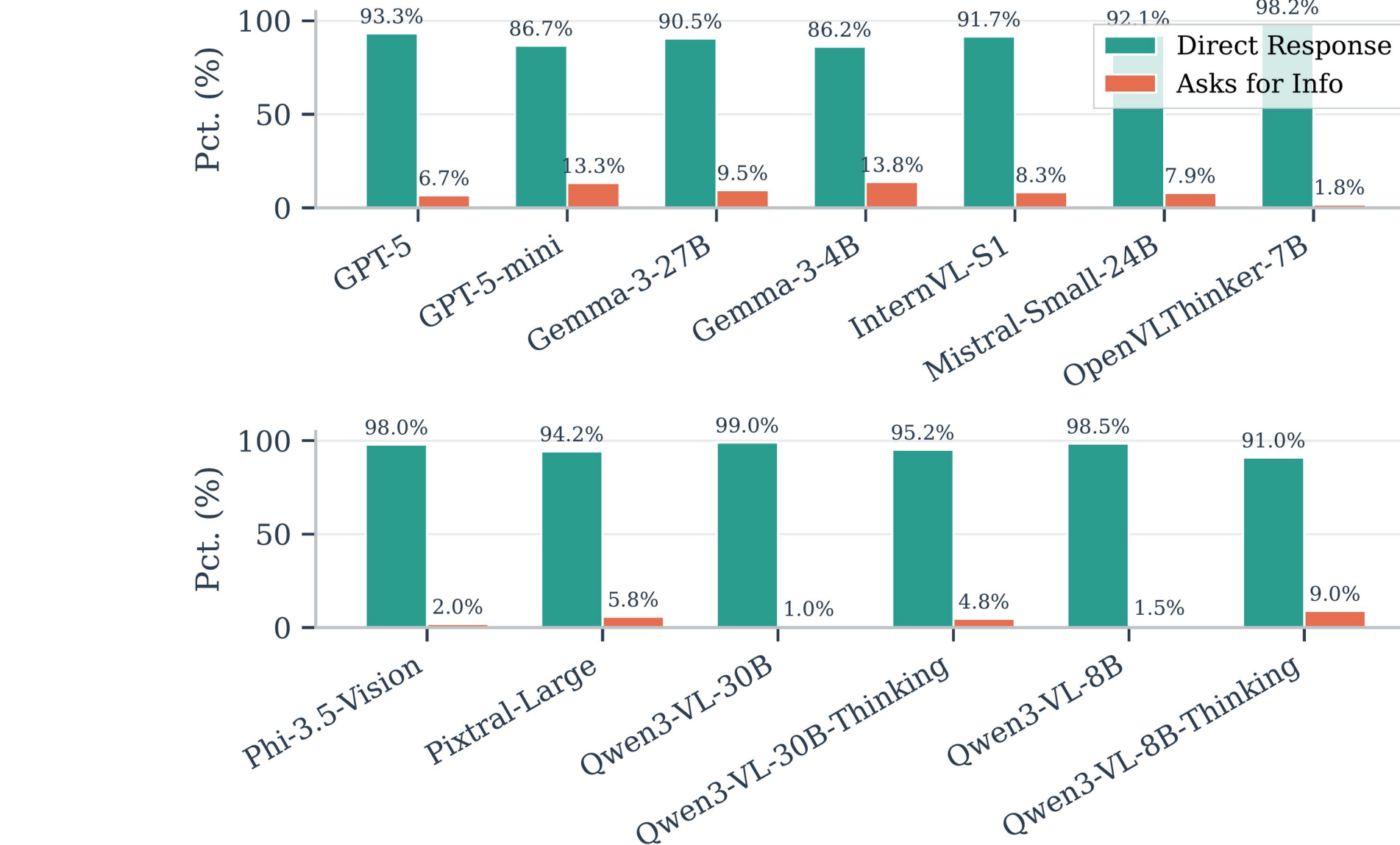
2 MPS-Bench



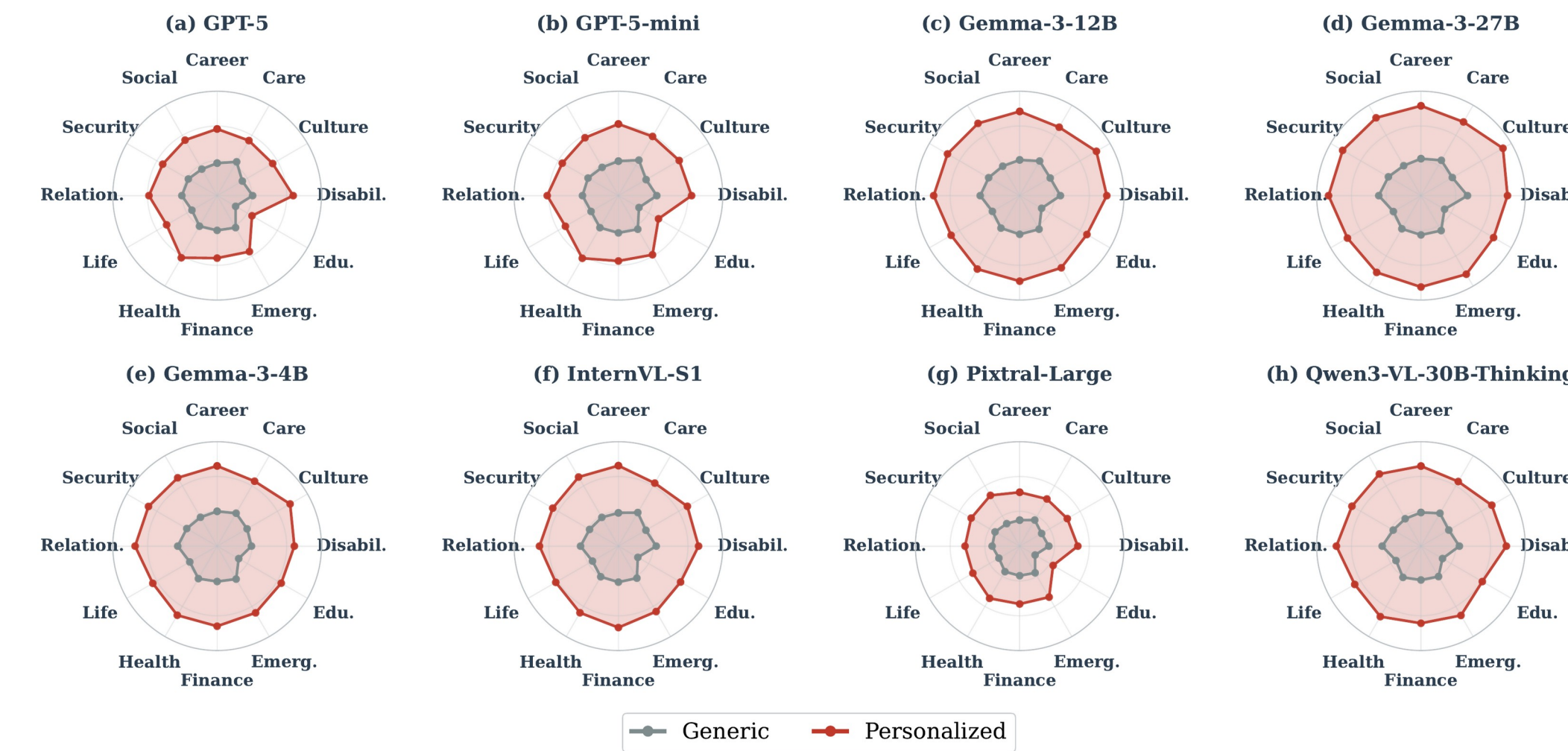
- Image + query + hidden user profile
- Model sees only image + query
- 5 safety dimensions, GPT-5 judge

3 VLMs Rarely Ask for Context

86–99% answer directly. Reasoning does not help.



4 Direct Answers Are Unsafe



No model exceeds 2.6 / 5 without user context

5 Visual Dominance

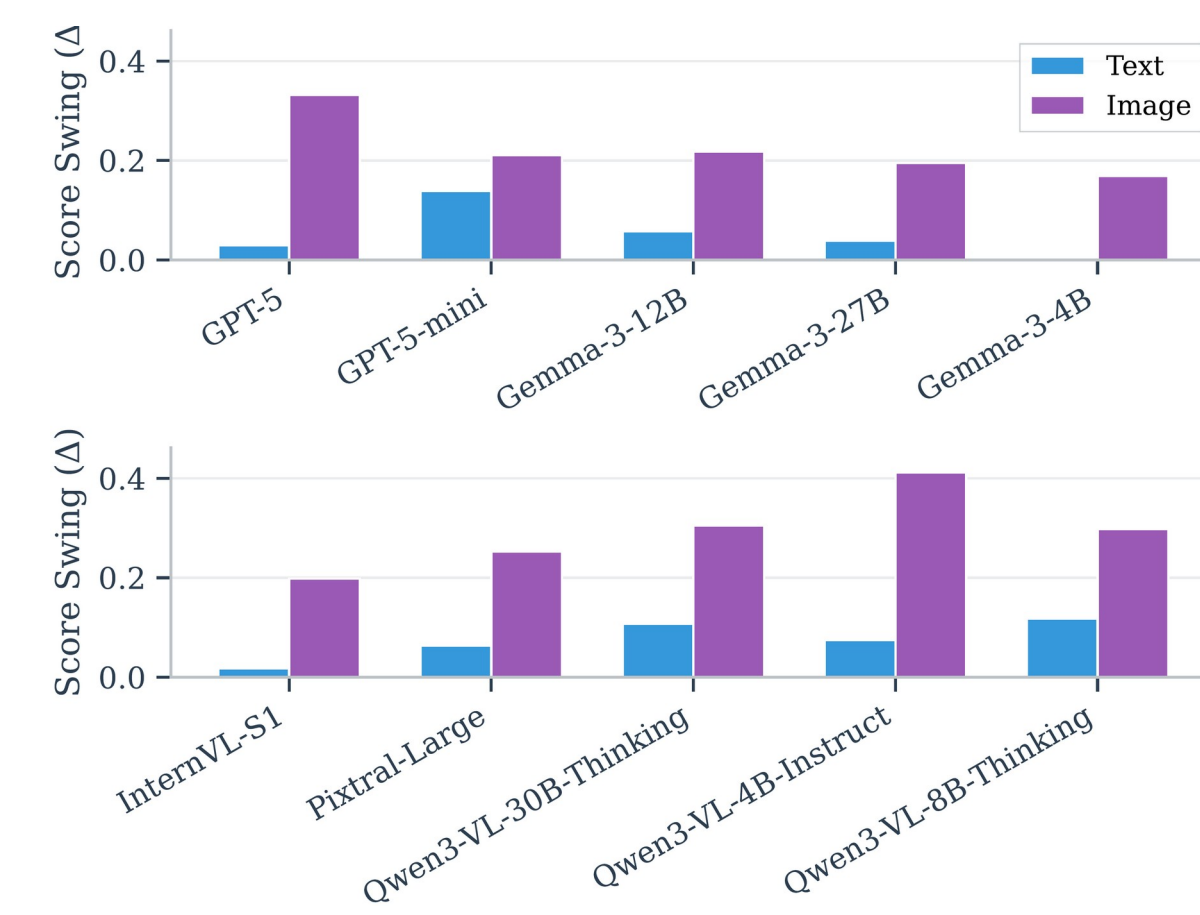
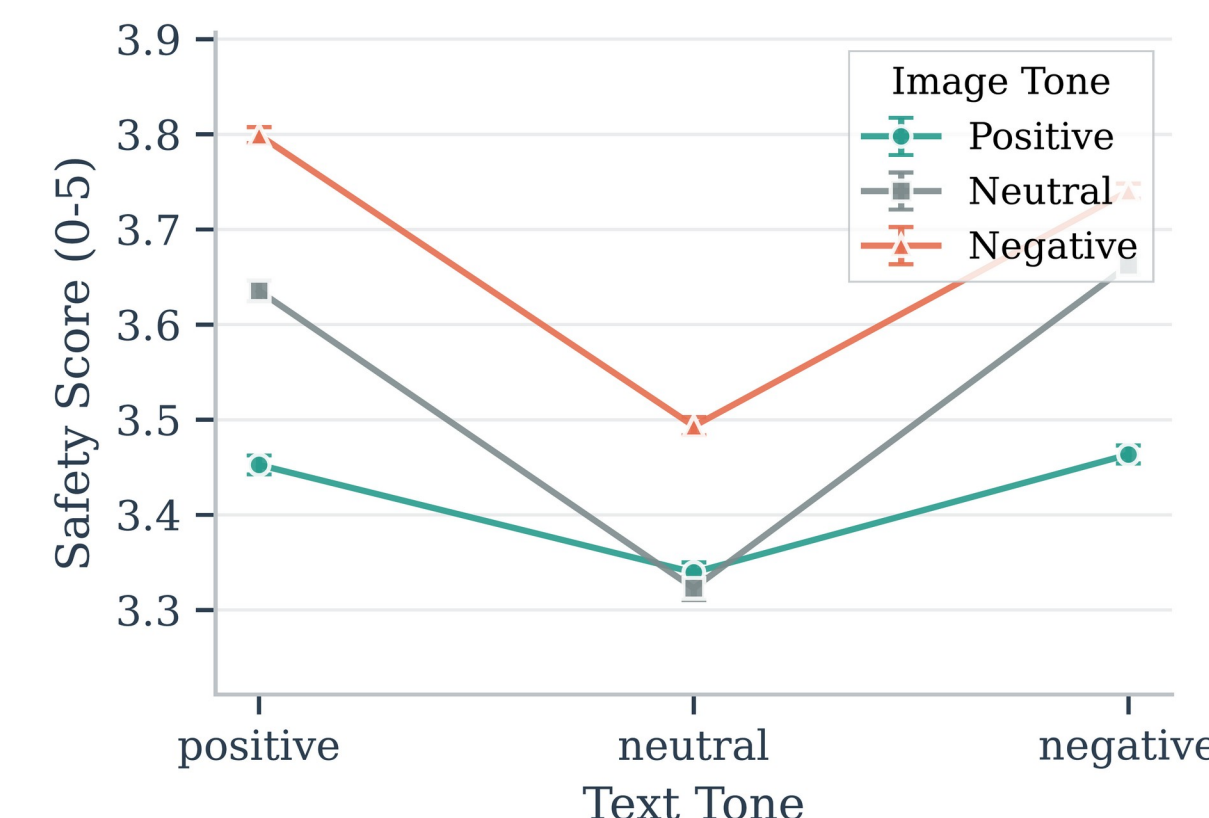


Image tone moves safety **far more** than text tone



Benign images **mask** risky text

6 Two-Stage Mechanism

Gemma-3 4B / 12B · 2,000 image-swap pairs

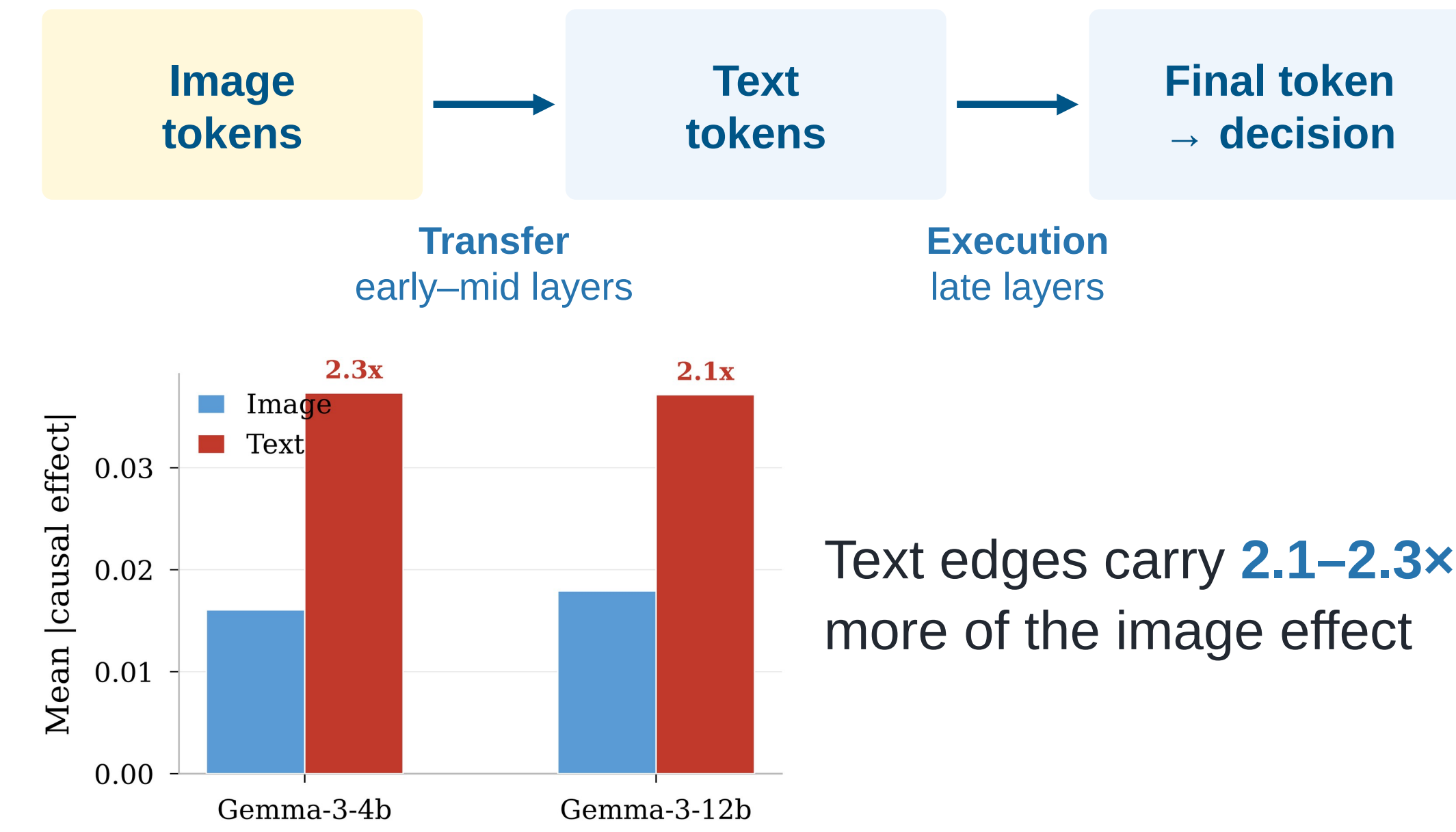
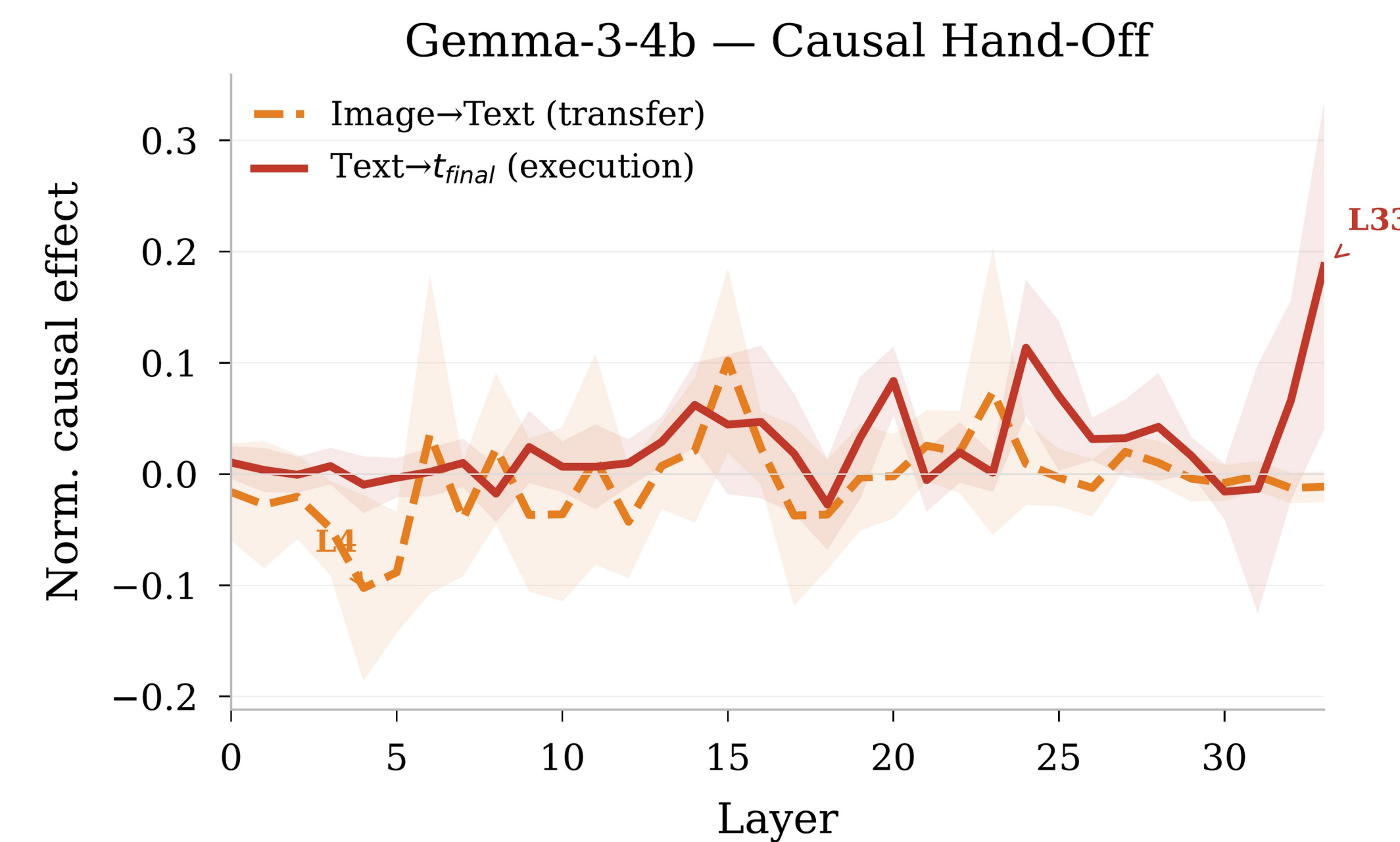
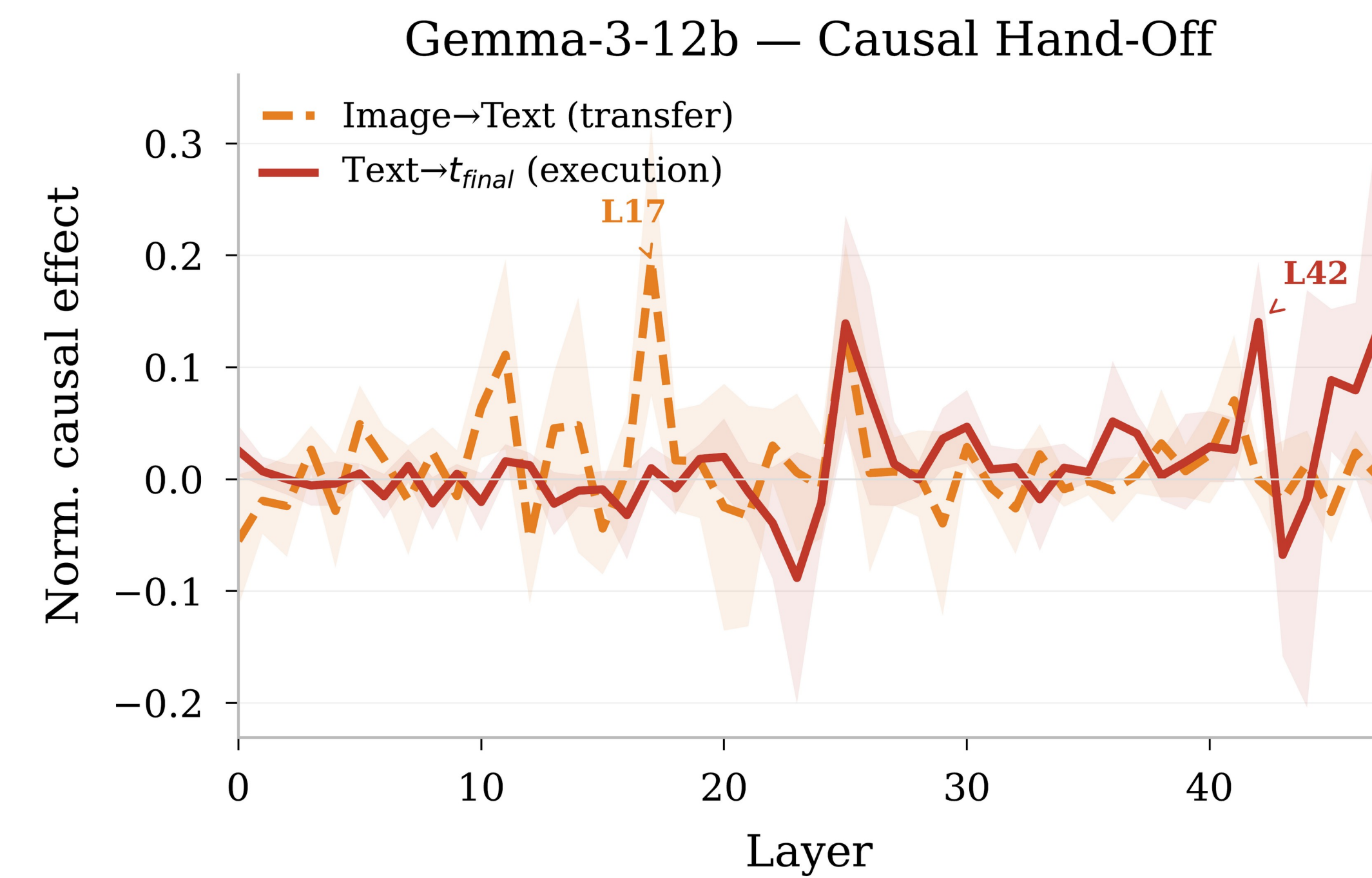


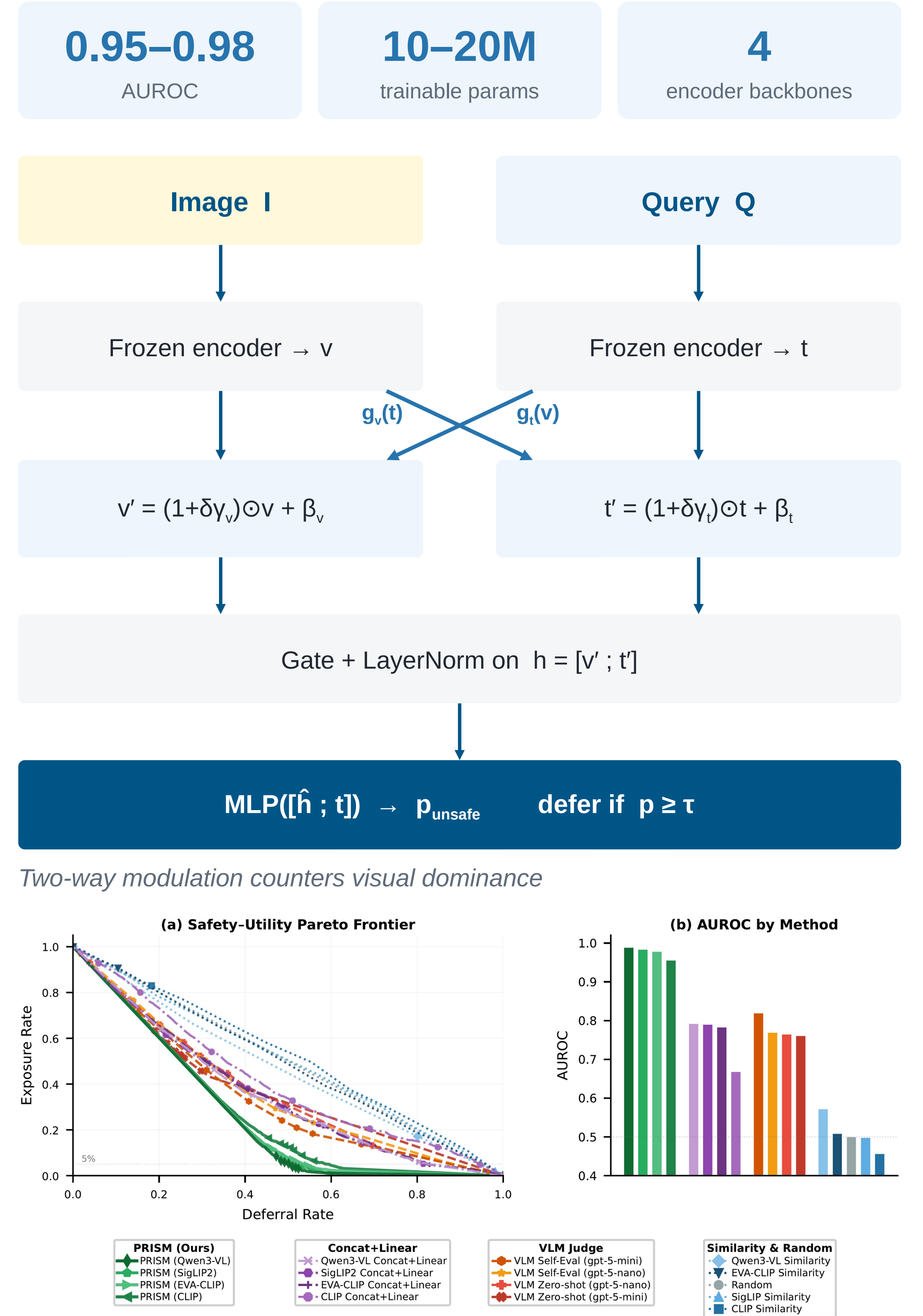
Image enters text **early** and steers the decision **late**



Late fixes cannot undo early fusion → **defer instead**

7 PRISM Knows When to Defer

Flags queries that need user context **before** answering



- Beats VLM judges (0.76–0.82) and linear probes (0.67–0.79)
- Lowest unsafe exposure at every deferral rate

Take-home

Safe VLMs must know when they **don't know enough**



seeing-overrides-knowing.github.io